

DATA ENGINEERING & ANALYTICS

The Global Listing Intelligence Initiative.

A Microsoft Fabric case study in wildcard ingestion, upsert integrity, audit trails, and pipeline orchestration for an ultra-luxury real estate firm operating across three currencies.

Grandeur Properties International

Ultra-luxury real estate · London · Dubai · New York
Data Engineering · Lakehouse · Governance · Audit

MICROSOFT FABRIC

CASE STUDY 01 · CONTENTS

Twelve sections. Five challenges. One pipeline that holds at seven.

A nightly listing feed from three offices, three currencies, and one head of portfolio who walks into the seven o'clock deal meeting needing the truth. Your job is to build the pipeline that gives it to her, every morning, without exception.

01

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SECTION 01 · EXECUTIVE RECOMMENDATION

Stop consolidating spreadsheets. Build the pipeline that decides for you.

Grandeur Properties International must transition immediately from manual, spreadsheet-based listing consolidation to an automated, auditable ingestion pipeline within Microsoft Fabric. The firm's expansion across London, New York, and Dubai has created a visibility gap that threatens both operational agility and portfolio governance.

THE FRAGILITY BEING NORMALISED

Analysts currently download nightly listing reports from each office and consolidate them into master spreadsheets. This introduces a latency of twenty four to forty eight hours in reporting, meaning leadership is making portfolio allocation decisions on stale data.

In the ultra-luxury market, where a single viewing converted to an offer can represent tens of millions of dollars, this delay is unacceptable. A modern firm cannot afford to walk into the seven o'clock meeting with yesterday's numbers.

WHAT LEADERSHIP MUST AUTHORISE

Deployment of a Global Listing Intelligence architecture that ingests, standardises, and secures nightly listing activity from all offices by 7:00 AM daily. This is not merely a technical upgrade. It is a strategic imperative to restore trust in the firm's intelligence capabilities.

"By decoupling data ingestion from manual intervention, leadership can act on high-value opportunities in near real time, securing Grandeur's position as the agile leader in ultra-luxury real estate."

KEY OUTCOMES

R

Reliability

Elimination of manual consolidation errors and missed file incidents via automated wildcard ingestion across all regional offices.

A

Accuracy

Upsert logic ensures corrected listing data from any office is reflected immediately in the Silver table without duplication.

C

Compliance

Full audit traceability through an immutable ingestion timestamp on every record, capturing exactly when data entered the pipeline.

S

Scalability

A plug-and-play architecture capable of onboarding new offices such as Singapore and Paris without pipeline restructure or code changes.

H

Hygiene

Automated archival and deletion of processed source files ensures the landing zone remains clean and prevents reprocessing of stale data.

G

Governance

PII columns explicitly excluded at the mapping stage. Personal contact details and internal system references never enter the governed layer.

SECTION 02 · VALUE AT STAKE

A thirty million pound penthouse, on yesterday's data.

The operational fragility of Grandeur's current data infrastructure presents a severe risk to both financial performance and competitive positioning. The simultaneous launch of three ultra-luxury listings across London, Dubai, and New York served as a definitive stress test that the manual process failed.

DIRECT FINANCIAL EXPOSURE

With properties listed at values exceeding ten million pounds, dollars, and dirhams, the inability to track viewings and offers in real time means the firm risks missing critical negotiation windows. A single missed offer signal on a thirty million pound London penthouse represents a commission loss that dwarfs the entire annual cost of a modern data platform.

During the March stress event, the firm was unable to report accurate listing activity for two consecutive days. Two portfolio decisions were made on incorrect data. The damage is not theoretical.

THE HIDDEN TAX ON THE ANALYST BENCH

Skilled analysts spend upwards of ten hours per week on low-value data consolidation. Downloading CSVs from three separate office systems, manually aligning columns, resolving version conflicts in shared spreadsheets. This work introduces duplicate records, missed file submissions, and silent overwrites.

The absence of any automated audit trail also means the firm cannot demonstrate data integrity to its institutional clients or legal counsel when disputes arise over offer timelines or valuation history.

01

Decision latency

Manual consolidation introduces a twenty four to forty eight hour delay between source files landing and leadership receiving a unified view of portfolio activity.

24-48h

Latency on portfolio intelligence

02

Analyst time absorbed

Hours per week burned on download, alignment, and reconciliation work that delivers no analytical value and produces errors at every handover.

10+ hrs

Weekly per analyst on consolidation

03

Audit gap

No ingestion timestamp on any record. Impossible to reconstruct the information landscape at the moment a portfolio decision was actually made.

0

Audit-ready timestamp coverage today

THE STRATEGIC FRAME

Every day leadership operates on manually assembled intelligence is a day where a competitor with a modern data estate holds a decisive advantage in deal velocity. The cost of the pipeline is trivial against the cost of one missed offer window.

SECTION 03 · INDUSTRY AND CLIENT CONTEXT

Belgravia, the Palm, and Central Park.

Founded in 1987, Grandeur Properties International represents buyers and sellers of the world's most exclusive residential assets. Georgian townhouses in Belgravia, sky villas on the Palm Jumeirah, and full-floor residences above Central Park. The firm trades on discretion, speed, and an unmatched read on the ultra-high-net-worth client.

THE OPERATING FOOTPRINT

Headquartered in Mayfair, London, with flagship offices in Dubai and New York. Grandeur operates at the intersection of wealth management and real estate, where relationships are everything and data integrity is non-negotiable.

Each office manages its own portfolio of active listings through a local CRM system, which exports a nightly snapshot as a CSV file. The file captures the current state of every active listing, including enquiries received, viewings scheduled, viewings completed, and whether an offer has been received.

THE INTELLIGENCE PROBLEM

Three files, one per office, sit in isolation every morning until a junior analyst manually downloads and consolidates them. By the time leadership receives a unified view, the data is already hours old and the morning deal meeting has begun without the full picture.

In an industry where the difference between closing a deal and losing it to a competitor is measured in hours, that gap is the entire problem.

EXHIBIT 01 · THE NIGHTLY LISTING INTELLIGENCE LANDSCAPE

OFFICE	CURRENCY	NIGHTLY FILE	CODE
London (HQ)	GBP	office_london_YYYY-MM-DD.csv · listing_price, enquiries_received, viewings_scheduled, viewings_completed, offer_received	LON
Dubai	AED	office_dubai_YYYY-MM-DD.csv · listing_price, enquiries_received, viewings_scheduled, viewings_completed, offer_received	DXB
New York	USD	office_newyork_YYYY-MM-DD.csv · listing_price, enquiries_received, viewings_scheduled, viewings_completed, offer_received	NYC

“

I need to walk into the seven o'clock deal meeting knowing exactly which properties have live offers, which had viewings last night, and which have gone cold. Right now I am walking in with yesterday's numbers at best. In this market, that is not good enough.

VICTORIA ASHWORTH · HEAD OF GLOBAL PORTFOLIO, GRANDEUR PROPERTIES INTERNATIONAL

SECTION 04 · PROBLEM DIAGNOSIS

A forty-eight hour window that broke three ways.

The fragility of Grandeur's manual model was exposed during a critical window when three major listings entered simultaneous offer negotiations. Dubai filed late. London duplicated rows. New York was silently overwritten. The morning deal meeting concluded, and two portfolio decisions were made on incorrect data.

EXHIBIT 02 · FAILURE MODE ANALYSIS

FAILURE MODE	WHAT HAPPENED	BUSINESS CONSEQUENCE
LATE FILE	Dubai CRM export delayed by two hours. No automated retry or alert mechanism existed.	Portfolio meeting proceeded without Dubai data. Offer status on Palm Jumeirah Sky Villa was unknown to leadership at the moment of decision.
MANUAL DUPLICATION	London analyst duplicated two rows during copy-paste consolidation into the master spreadsheet.	London viewing count artificially inflated. Capital allocation decision skewed by an engagement signal that did not exist.
VERSION CONFLICT	New York file overwritten in shared drive by an older version. No version control or immutable storage in place.	New York listing status reverted to the previous day's data. No one noticed until after the meeting had ended.
NO AUDIT TRAIL	No ingestion timestamp on any record. Impossible to determine when data entered the system.	Firm unable to reconstruct the information landscape at the time decisions were made. Legal and compliance exposure.

This sequence confirms three distinct and compounding weaknesses. There is no automated mechanism to detect a late or missing file. There is no row-level integrity check to identify duplicates introduced during manual handling. There is no audit trail to determine when data entered the system. Without intervention, each weakness scales directly with the number of offices. Singapore and Paris will add two more daily files to a process already at the edge of failure.

DIAGNOSTIC FINDING

The current dependence on manual file consolidation creates an unacceptable latency of twenty four to forty eight hours and a high probability of material error at every stage. Without an automated ingestion pipeline that captures a clean nightly snapshot, enforces upsert integrity on property-level records, archives processed files, and timestamps every record at ingestion, Grandeur Properties International cannot guarantee the accuracy of the intelligence its leadership relies on every morning.

SECTION 05 · TARGET OPERATING MODEL

From manual consolidation to orchestrated trust.

Grandeur will transition to a Lakehouse-centric architecture within Microsoft Fabric. Data Engineering assumes full accountability for the automated ingestion, validation, and archival of nightly listing files. Raw file handling is decoupled from business consumption, so leadership trusts the data without needing to understand how it was assembled.

The core shift is from manual consolidation to automated orchestration through a three-activity Microsoft Fabric Pipeline. The first activity ingests all nightly office files using a wildcard pattern, maps defined columns to a governed Silver table, and upserts each record using `property_id` as the unique key. The second archives processed files. The third deletes them from the landing zone. Three steps, one schedule, every morning.

EXHIBIT 03 · STRATEGIC SHIFT IN OPERATING MODEL

DIMENSION	CURRENT STATE	FUTURE STATE
Process ownership	Junior analysts manually download, align, and consolidate three office files every morning.	Microsoft Fabric Pipeline automatically ingests all office files via wildcard at 6:00 AM UTC daily.
Data integrity	No duplicate detection. Copy-paste errors go undetected until a downstream decision is questioned.	Upsert on <code>property_id</code> ensures one authoritative record per property. Corrections overwrite, never duplicate.
Audit trail	No ingestion timestamp. Impossible to determine when data entered the system or reconstruct past states.	Every record stamped with a UTC ingestion timestamp at the moment the pipeline processes the file.
File hygiene	Processed files remain in shared drives indefinitely, creating version conflicts and reprocessing risk.	Processed files automatically archived then deleted from landing zone after every successful pipeline run.
Scalability	Each new office requires a new manual download step and a new analyst trained on consolidation.	New offices are onboarded by dropping files into the landing folder. Zero pipeline changes required.

SECTION 06 · ENGINEERING MANDATE

Four non-negotiable constraints. One pipeline.

You are the Lead Data Engineer at Grandeur Properties International, reporting to the Head of Global Portfolio. Your mandate is to design and operationalise the Global Listing Intelligence pipeline within Microsoft Fabric. The architecture is governed by four constraints that define every design decision you will make.

EXHIBIT 04 · ENGINEERING CONSTRAINTS AND JUSTIFICATIONS

CONSTRAINT	CONFIGURATION	JUSTIFICATION
Wildcard Ingestion	Files/raw/ · wildcard <code>office_*.csv</code> . Captures all current and future office files without hardcoding.	Zero-touch onboarding
Upsert Write Mode	Write behaviour Upsert , key column <code>property_id</code> . One authoritative record per property.	No duplicates
Ingestion Timestamp	Additional column <code>ingestion_timestamp</code> with value <code>@utcnow()</code> . Immutable record of when data entered the layer.	Audit anchor
PII Exclusion	Excluded from mapping: <code>agent_personal_email</code> and <code>internal_crm_ref</code> . Never enter the Silver layer.	Privacy by design
Archive Activity	Source <code>Files/raw/office_*.csv</code> , destination <code>Files/archive/</code> . Immutable historical record.	Recoverability
Delete Activity	Target <code>Files/raw/office_*.csv</code> . Keeps the landing zone clean and prevents reprocessing of stale data.	Hygiene

The pipeline must be simple enough to be maintained by a single engineer and scalable enough to onboard two additional offices without structural changes. Hardcoding individual filenames is prohibited. As new offices come online, their files must be picked up automatically. The dynamic expression `@utcnow()` is the firm's audit mechanism and must be present on every row without exception. The two PII columns carry no analytical value and their presence in a governed data layer represents an unnecessary privacy risk.

EXHIBIT 05 · SILVER TABLE TARGET SCHEMA

silver_listing_pipeline.

Twelve mapped columns, one pipeline-generated audit field, two columns explicitly excluded. The shape of the table is the shape of the contract you are signing with the Head of Global Portfolio.

COLUMN	TYPE	NOTES	ROLE
property_id	String	Unique identifier per property. Used as the upsert key.	UPSERT KEY
property_name	String	Human-readable listing name.	Mapped
listing_price	Decimal	In local office currency.	Mapped
currency	String	ISO currency code. GBP, AED, or USD.	Mapped
enquiries_received	Integer	Cumulative enquiries since listing went live.	Mapped
viewings_scheduled	Integer	Total viewings currently scheduled.	Mapped
viewings_completed	Integer	Total viewings completed to date.	Mapped
offer_received	String	Y or N. Triggers immediate leadership attention.	Mapped
last_refreshed	Date	Date the CRM exported this record.	Mapped
office_code	String	Source office identifier. LON, DXB, or NYC.	Mapped
agent_id	String	Responsible agent identifier.	Mapped
ingestion_timestamp	DateTime	Pipeline-generated. UTC timestamp added at execution time via @utcnow().	AUDIT
agent_personal_email	EXCLUDED	PII. Must not appear in Silver table under any circumstance.	DROP
internal_crm_ref	EXCLUDED	System artefact. Must not appear in Silver table under any circumstance.	DROP

SECTION 07 · DELIVERY ROADMAP

Three phases. One pipeline. Built in deliberate sequence.

Risk is managed by introducing capability in stages rather than deploying everything at once. Foundation first. Integrity second. Production scheduling last. Each phase produces a verifiable outcome before the next is allowed to begin.

Phase 01 Foundation

Ingest and Validate

Deploy the Copy Data activity with wildcard ingestion, column mapping, PII exclusion, upsert write mode, and ingestion timestamp. Run the pipeline manually against the Day 1 files for all three offices. Validate row counts in the Silver table and confirm the

```
ingestion_timestamp
```

column is populated on every record.

Success: 9 clean records in Silver. No PII columns present.

Phase 02 Integrity

Upsert Verification

Replace the Day 1 files in

```
Files/raw/
```

with the Day 2 files. Run the pipeline manually again. Verify that updated records reflect new values from Day 2 and that the total row count remains at nine. No new rows should be inserted. Only changed fields should reflect new values.

Success: Confirmed upsert behaviour with zero duplicate records.

Phase 03 Production

Archive, Delete and Schedule

Add the Archive activity to copy all processed files from

```
Files/raw/
```

to

```
Files/archive/
```

on success. Add the Delete activity to remove all files from

```
Files/raw/
```

after archival is confirmed. Schedule the pipeline to execute daily at 6:00 AM UTC.

Success: Raw is empty. Archive is populated. Silver reflects the latest state.

SECTION 08 · ACCEPTANCE CRITERIA

Seven checks. All must pass.

All criteria must be satisfied before the pipeline is considered production-ready. Each is verified through Microsoft Fabric Pipeline Monitoring and direct SQL queries against the Silver table via the Lakehouse SQL endpoint.

01	Completeness	The pipeline must identify and ingest all office files matching the <code>office_*.csv</code> wildcard pattern from <code>Files/raw/</code> without manual intervention.
02	Integrity	Total row count in the Silver table must exactly match the total number of unique property records across all source files. No duplicates permitted.
03	Upsert Behaviour	Running the pipeline with Day 2 files must update changed fields on existing records without inserting new rows for already-existing property identifiers.
04	Audit Trail	Every record in the Silver table must carry a valid UTC <code>ingestion_timestamp</code> generated at pipeline execution time. No null timestamps permitted.
05	PII Exclusion	Columns <code>agent_personal_email</code> and <code>internal_crm_ref</code> must be strictly absent from the Silver table under all circumstances.
06	Archival	After every successful pipeline run, all processed source files must be present in <code>Files/archive/</code> and absent from <code>Files/raw/</code> .
07	Scheduling	The pipeline must execute automatically at 6:00 AM UTC daily without manual trigger.

VALIDATION METHOD

Each criterion is verified through a combination of the Microsoft Fabric Pipeline Monitoring view, which confirms activity-level success and rows read versus rows written, and direct SQL queries against the Silver table via the Lakehouse SQL endpoint to confirm row counts, column presence, timestamp validity, and the absence of PII columns.

SECTION 09 · RISKS AND MITIGATIONS

Where this pipeline could still hurt you.

Every automated pipeline trades one class of risk for another. Manual errors fall away. New ones replace them. The list below is what you must be ready to answer for when a senior engineer asks where this design is fragile.

CATEGORY	SPECIFIC RISK	MITIGATION STRATEGY
TECHNICAL	Schema drift. An office CRM vendor renames a column without notifying the Data Engineering team.	Explicit column mapping in the Copy Data activity means only defined columns enter the Silver table. An unmapped column is silently dropped. A renamed key column such as <code>offer_received</code> will result in a null value in Silver, which is detectable via a post-run row validation query.
OPERATIONAL	Missing file. An office fails to submit their nightly file before the 6:00 AM UTC pipeline trigger.	The wildcard pattern picks up whatever files are present. The pipeline succeeds on available files. Properties from the missing office retain their previous values in the Silver table. Leadership must be made aware that absence of an update does not mean absence of activity.
DATA QUALITY	Duplicate source records. An office CRM exports the same property twice in a single nightly file.	Upsert logic on <code>property_id</code> processes each row sequentially. The last occurrence of a duplicate <code>property_id</code> in the source file wins. This is acceptable for a nightly snapshot model where the final state of the day is the authoritative record.
PIPELINE FAILURE	Archive activity fails after Copy Data succeeds. Delete activity does not execute.	Source files remain in <code>Files/raw/</code> . On the next pipeline run, the Copy Data activity reprocesses them. Upsert logic ensures no duplicates in Silver. The archive gap is an operational issue, not a data integrity issue. Pipeline dependency configuration ensures Delete never runs without a confirmed Archive success.

SECTION 10 · DISCUSSION FRAMEWORK

The questions behind the configuration.

These are not comprehension checks. They are the questions a senior engineer or hiring manager would ask you to test whether you understand the reasoning behind the decisions, not just the steps you followed to implement them.

Wildcard Design	The pipeline uses <code>office_*.csv</code> as the wildcard pattern rather than <code>*.csv</code> . Why is this distinction important? What could go wrong if the pattern were too broad?
Upsert versus Append	The write mode is configured as Upsert rather than Append. What would happen to the Silver table over time if Append were used instead? Under what business circumstance would Append actually be the correct choice?
Ingestion Timestamp Purpose	The <code>ingestion_timestamp</code> records when the pipeline processed the file, not when the data was generated by the CRM. What is the practical difference between these two timestamps and why does that distinction matter for audit purposes?
Archive before Delete	The pipeline is configured to archive files before deleting them, with Delete dependent on Archive success. Why is this sequencing non-negotiable? What would be the consequence of reversing the order?
PII Exclusion Strategy	The column mapping explicitly excludes <code>agent_personal_email</code> and <code>internal_crm_ref</code> . What happens to these columns when the pipeline runs? Where do they go? Is exclusion from mapping sufficient as a privacy control, or are there additional measures a production environment would require?

SECTION 11 · LAB INSTRUCTIONS

Build the pipeline. Victoria walks in at seven.

You are the Data Engineer at Grandeur Properties International. Victoria Ashworth walks into the seven o'clock deal meeting every morning needing one thing: a single, accurate, current view of every active listing across London, Dubai, and New York. Right now she does not have it. You are going to build the pipeline that gives it to her.

TWO WAYS TO SOLVE THIS PROBLEM

Before building anything, consider how this problem could be approached and why one approach is clearly superior for a firm operating at this scale.

DIMENSION	APPROACH A · MANUAL CONSOLIDATION	APPROACH B · FABRIC PIPELINE
Execution	Analyst downloads and consolidates files each morning.	Pipeline fires automatically at the same time every day.
Consistency	Depends entirely on analyst attention and process discipline.	Identical behaviour on every run without variation.
Audit trail	No record of when data entered the system or who handled it.	Ingestion timestamp stamped on every record at execution time.
PII control	Analyst must remember to remove restricted columns each time.	Excluded columns never reach the Silver table under any circumstance.
Error recovery	Errors discovered after decisions have already been made.	Pipeline fails explicitly. Monitoring identifies the exact point of failure.
Scalability	Each new office adds a new manual step to an already fragile process.	New offices are onboarded automatically. No pipeline changes required.
RECOMMENDATION	Not suitable for production use at this scale.	Recommended. This is what you will build.

YOUR SOURCE FILES

FILE	OFFICE	PURPOSE
office_london_2026_04_03.csv	LONDON	Day 1 initial load
office_dubai_2026_04_03.csv	DUBAI	Day 1 initial load
office_newyork_2026_04_03.csv	NEW YORK	Day 1 initial load
office_london_2026_04_04.csv	LONDON	Day 2 upsert validation
office_dubai_2026_04_04.csv	DUBAI	Day 2 upsert validation
office_newyork_2026_04_04.csv	NEW YORK	Day 2 upsert validation

YOUR MANDATE · BUILD, VALIDATE, VERIFY

Pipeline architecture and validation gates.

Build a Microsoft Fabric Pipeline that satisfies every engineering constraint and produces the architecture below. The pipeline must be fully autonomous, leaving no manual steps between the arrival of office files and the availability of a clean, governed Silver table by 7:00 AM every morning.

VALIDATE DAY 1 · INGESTION

Validation Check	Expected Result
Total row count	9 rows. Three per office.
ingestion_timestamp column	Present on every row. No null values.
agent_personal_email	Column must not exist in the table.
internal_crm_ref	Column must not exist in the table.
property_id values	Nine unique values. No duplicates.

VALIDATE DAY 2 · UPSERT BEHAVIOUR

Validation Check	Expected Result
Total row count	Still 9 rows. No new rows inserted.
GB-BLG-19001 offer_received	Updated to Y. Was N in Day 1.
AE-PLM-18901 viewings_completed	Incremented from Day 1 value.
US-PKA-35001 listing_price	Reflects updated Day 2 price.
ingestion_timestamp	Updated to reflect the Day 2 pipeline run time.

EXHIBIT 06 · FINAL PIPELINE ARCHITECTURE

#	Activity	Type	Depends On	Purpose
01	ACT_IngestListingFiles	Copy Data	First activity	Wildcard ingest from landing zone, upsert to silver_listing_pipeline, add ingestion_timestamp, exclude PII columns.
02	ACT_ArchiveListingFiles	Copy Data	ACT_IngestListingFiles On Success	Copy all processed files from landing zone to archive folder for immutable historical record.
03	ACT_DeleteProcessedFiles	Delete	ACT_ArchiveListingFiles On Success	Remove all processed files from landing zone to keep it clean for the next pipeline run.

SECTION 12 · PIPELINE DESIGN CHALLENGES

Five scenarios. No step-by-step. Defensible reasoning.

You have successfully built the Grandeur Properties Global Listing Intelligence pipeline. What follows are five real-world scenarios that Data Engineers encounter after a pipeline goes live. There are no step-by-step instructions. You are expected to reason through each situation, arrive at a defensible design decision, and articulate your thinking clearly. These challenges are portfolio-ready. Document your approach as if presenting to a senior architect or a hiring manager.

CHALLENGE 01

The Singapore Expansion

A new office begins dropping files into the landing zone next Monday. Does the pipeline need to be modified before then? You have not changed it since go-live.

TESTS · WILDCARD REASONING

CHALLENGE 02

The Late Correction

Dubai contacts you at 11:30 PM to report an incorrect listing price already sitting in Silver. They drop a corrected file with a different filename. What happens when you run it?

TESTS · UPSERT IDEMPOTENCY

CHALLENGE 03

The Vanishing Column

A vendor renames a key column overnight. The pipeline fails on the New York file. London and Dubai succeed. What is the minimum change that resolves it without breaking the others?

TESTS · SCHEMA DRIFT

CHALLENGE 04

The Broken Chain

Ingest succeeds. Archive fails. Delete does not execute. The next morning brings new files alongside the unprocessed ones. Walk through every activity's behaviour.

TESTS · DEPENDENCY DESIGN

CHALLENGE 05

The Missing Office

London never submits. The pipeline still completes with Succeeded status. What does Victoria see on her dashboard at 7:00 AM, and what could she incorrectly conclude? This one exposes a fundamental limitation in the design.

TESTS · SILENT FAILURE DETECTION · FUTURE-STATE THINKING

PIPELINE DESIGN CHALLENGE 01

01.

The Singapore Expansion

Grandeur Properties International has signed a lease for a new office in Singapore. The Singapore team will begin dropping their nightly listing file into `Files/raw/` from next Monday. Their file is named `office_singapore_2026_04_07.csv` and follows the identical schema and naming convention as the London, Dubai, and New York files. You have not modified the pipeline since go-live.

YOUR CHALLENGE

What happens on Monday morning when the pipeline fires? Does the pipeline need to be modified before Monday? If yes, what specifically needs to change and why? If no, explain precisely why the current configuration handles the new office without intervention. What would you check in the pipeline Monitoring view after the first Singapore run to confirm everything worked as expected?

SAMPLE ARTEFACT PROVIDED

A sample Singapore file `office_singapore_2026_04_07.csv` has been provided with this case study for you to test your answer end to end.

PIPELINE DESIGN CHALLENGE 02

02.

The Late Correction

At 11:30 PM on a Thursday, the Dubai office contacts you to report that their nightly file contained an incorrect `listing_price` for property `AE-PLM-18901`. The correct price is 19,500,000 AED, not the 18,900,000 AED that was submitted. The pipeline already ran successfully at 6:00 AM that morning and the incorrect value is currently sitting in the Silver table. The Dubai team have placed a corrected file named `office_dubai_2026_04_03_corrected.csv` into `Files/raw/` and are asking you to reprocess it immediately.

YOUR CHALLENGE

Can the pipeline handle this reprocessing request without modification? Walk through exactly what happens to the existing AE-PLM-18901 record in the Silver table when the pipeline runs against the corrected file. Does the filename matter? What happens to the corrected file after the pipeline completes? What would you verify in the Silver table to confirm the correction landed correctly? Is there anything about this scenario that concerns you from a data governance perspective?

SAMPLE ARTEFACT PROVIDED

A sample corrected file `office_dubai_2026_04_03_corrected.csv` has been provided with this case study so you can validate the reprocessing path end to end.

PIPELINE DESIGN CHALLENGE 03

03.

The Vanishing Column

On a Tuesday morning the pipeline fails. Investigating the error log you discover that the New York office CRM vendor pushed an overnight software update that renamed the column

`offer_received` to `offer_status` in all exported files. The London and Dubai files processed correctly. Only the New York file caused the failure.

YOUR CHALLENGE

Explain exactly why the pipeline failed at this specific point. What does the pipeline see when it encounters `offer_status` in place of `offer_received`? What is the current state of the Silver table for New York properties after this failure? What is the minimum change required to resolve this without breaking the London and Dubai mappings? How would you prevent this class of failure from causing a silent data quality issue rather than an explicit pipeline failure in the future?

SAMPLE ARTEFACT PROVIDED

A sample New York schema drift file `office_newyork_schema_drift_2026_04_03.csv` has been provided with this case study to help you investigate the failure mode.

PIPELINE DESIGN CHALLENGE 04

04.

The Broken Chain

On a Wednesday morning the pipeline runs and `ACT_IngestListingFiles` completes successfully, upserting all nine property records into the Silver table. However, `ACT_ArchiveListingFiles` fails halfway through due to a transient storage connectivity issue. As a result, `ACT_DeleteProcessedFiles` does not execute. The three source files remain in `Files/raw/`. The pipeline is shown as Failed in the Monitoring view.

YOUR CHALLENGE

What is the current state of the Silver table? What is the current state of `Files/raw/` and `Files/archive/`? Is the data in the Silver table trustworthy? When the pipeline runs again the following morning with new office files also present in `Files/raw/`, what happens? Walk through the behaviour of all three activities given that both Wednesday and Thursday files are now in the landing zone. Is the On Success dependency configuration between Archive and Delete doing its job in this scenario? What does this tell you about the importance of pipeline dependency design?

PIPELINE DESIGN CHALLENGE 05

05.

The Missing Office

On a Friday morning the London office CRM system crashes at 11:00 PM the previous night and never completes its nightly export. The Dubai and New York files land in `Files/raw/` on time. The London file never arrives. The pipeline fires at 6:00 AM UTC as scheduled and completes with a **Succeeded** status.

YOUR CHALLENGE

Why did the pipeline succeed despite a missing file? What does the wildcard pattern do when only two of the three expected files are present? What is the current state of the London property records in the Silver table? When Victoria Ashworth opens the dashboard at 7:00 AM, what does she see for London listings and what could she incorrectly conclude? What is the fundamental limitation of the current pipeline design that this scenario exposes, and what would a more mature architecture include to address it? Note that the solution to this limitation is outside the scope of this case study but is worth articulating as a future design consideration.

FORWARD-LOOKING NOTE

This is the only challenge where the answer points *beyond* what this pipeline does. Use it to demonstrate that you understand where the current design ends and where the next maturity step begins.

From three files at 6:00 AM to one trusted view at seven.

Build the pipeline that gives Victoria Ashworth the truth every morning. Wildcard ingestion, upsert integrity, timestamped audit, PII exclusion, immutable archive, automatic cleanup. Six engineering decisions, one architecture, one Silver table that leadership can finally trust.

WHAT YOU WILL DELIVER

A Microsoft Fabric Pipeline composed of three orchestrated activities, scheduled at 6:00 AM UTC, writing to a governed Silver table named `silver_listing_pipeline`.

HOW YOU WILL BE EVALUATED

Seven acceptance criteria, two validation gates across Day 1 and Day 2, five real-world challenges that test reasoning rather than recall.

WHAT THIS PROVES

You can take a vague business pain, reduce it to a small number of engineering constraints, and deliver a pipeline that fails explicitly when something is wrong and succeeds quietly when everything is right.

WHAT COMES NEXT

Onboard Singapore. Onboard Paris. Add a missing-file alert. Add a row-count validation step. Move the Silver table into a wider semantic layer. Each is a small change. The pipeline is the platform.

INSTRUCTOR CONTACT

Surya Kumar · suryakumar01985@gmail.com